

KRRISH BHASIN

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EDUCATION

BITS Pilani, K.K. Birla Goa Campus

Goa, India | Expected May 2028

B.E. (Hons.) Computer Science

- Relevant coursework: Linear Algebra, Probability & Statistics, Foundations of Data Science, Machine Learning, OOP
- Achievement: Codeforces Specialist (peak rating 1546)

EXPERIENCE

ML Research Intern, Indian Institute of Technology Bombay

Mumbai | Aug 2025 - Feb 2026

Natural Language Processing / Machine Translation Lab, advised by Prof. Ganesh Ramakrishnan

- Researched neural machine translation for Indic languages with transformer models (IndicTrans2) across Hindi-English and multi-Indic pairs.
- Fine-tuned models in Fairseq on large parallel corpora; benchmarked quality with BLEU and chrF across language pairs.
- Built an LLM post-correction pipeline with domain-specific glossary injection, improving translation consistency for technical and rare-domain vocabulary.
- Contributed to the Udaan OCR/MT pipeline for digitizing Indian-language educational content; diagnosed tokenization mismatches and stabilized training runs.
- Ran controlled ablations comparing fine-tuned vs. zero-shot performance and presented findings to the lab.

Software Engineering Intern, Hodo Stays

Bengaluru | May 2025 - Jul 2025

LLM-powered guest communication for a hospitality SaaS platform

- Designed and shipped a multi-channel (WhatsApp, email, in-app) guest messaging system that automated routine guest communication.
- Engineered a three-class intent classifier (booking / maintenance / general) reaching ~91% accuracy on an internal test set, cutting manual triage.
- Designed prompt templates with dynamic context injection (guest name, property, booking status) to produce brand-aligned, grounded responses.
- Built the backend on Supabase (PostgreSQL, real-time subscriptions) and routed model calls through OpenRouter, keeping median response time around 1.4s.

PROJECTS

Improving NER with a Mixture-of-Experts Layer

github.com/Levi0p/moe-ner

Python, PyTorch, HuggingFace

- Implemented a Mixture-of-Experts layer from scratch (4 expert MLPs, top-2 gating, 4.2M params vs. 3.7M baseline) and integrated it into a BiLSTM tagger for named-entity recognition.
- Designed the MoE block with differentiable top-2 routing and a residual + LayerNorm connection to prevent representation collapse.
- Ran ablation studies on CoNLL-2003: entity-F1 improved from 71.90% to 72.89%, while accuracy dipped slightly by 0.32%.

Instruction-Tuned LLM for Domain-Specific QA

github.com/Levi0p/Instruction-Tuned-LLM-for-Domain-Specific-QA

Python, PyTorch, HuggingFace, FAISS

- Fine-tuned LLaMA with LoRA (parameter-efficient fine-tuning) on a curated domain-specific instruction dataset.
- Added a FAISS retrieval-augmented pipeline over the domain corpus to reduce hallucination and improve factual grounding.
- Evaluated with BLEU and ROUGE; LoRA + RAG outperformed zero-shot prompting on answer relevance and factual accuracy.

CO-CURRICULAR

AlgoManiax, Competitive Programming Club, BITS Goa (Core Member, Apr 2025 - Present)

- Mentor peers in competitive programming and organize ICPC-format mock contests and team practice sessions.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, SQL, Bash

ML / AI: PyTorch, TensorFlow, HuggingFace Transformers, Fairseq, LoRA / PEFT, Scikit-learn, NumPy

LLM & NLP: RAG pipelines, LangChain, FAISS, prompt engineering, IndicTrans2, BLEU / chrF / ROUGE evaluation

Systems & Web: React, Flask, REST APIs, Supabase (PostgreSQL), Git, Linux, OpenCV